

FAULT CODE 123 - Intake Manifold 1 Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement; Part Number 3164596 - male Framatome™ test lead and Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary		
STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for sensor supply fault codes.	Sensor supply fault codes active?
	STEP 1B. Check for an inactive fault code.	Fault Code 123 inactive?
STEP 2.	Check the intake manifold pressure sensor and circuit.	
	STEP 2A. Inspect the intake manifold pressure sensor and connector pins.	Dirty or damaged pins?
	STEP 2B. Check the sensor supply voltage and return circuit.	Between 4.75-VDC and 5.25-VDC?
	STEP 2C. Check the circuit response.	Fault Code 122 active and Fault Code 123 inactive?

STEPS		SPECIFICATIONS
	STEP 2D. Check the fault codes and verify sensor condition.	Fault Code 123 active?
STEP 3.	Check the ECM and engine harness.	
	STEP 3A. Inspect the ECM and engine harness connector pins.	Dirty or damaged pins?
	STEP 3A-1. Check the sensor supply voltage and return circuit.	Between 4.75-VDC and 5.25-VDC?
	STEP 3A-2. Check for an active fault code.	Fault Code 123 inactive?
	STEP 3B. Inspect the ECM and engine harness connector pins.	Dirty or damaged pins?
	STEP 3B-1. Check the ECM response.	Fault Code 122 active and Fault Code 123 inactive?
	STEP 3B-2. Check for an active fault code.	Fault Code 123 inactive?
STEP 4.	Check ECM calibration and clear fault codes.	
	STEP 4A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 4B. Disable the fault code.	Fault code inactive?

STEP 1. Check the fault codes.

STEP 1A. Check for sensor supply fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for sensor supply fault codes. Use INSITE™ electronic service tool to read the fault codes.	Sensor supply fault codes active? YES	Appropriate troubleshooting symptom tree
	Sensor supply fault codes active? NO	1B

STEP 1B. Check for an inactive fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for inactive fault code. Use INSITE™ electronic service tool to read the inactive fault codes.	Fault Code 123 inactive? YES	Use the following procedure for inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
	Fault Code 123 inactive? NO	2A

STEP 2. Check the intake manifold pressure sensor and circuit.

STEP 2A. Inspect the intake manifold pressure sensor and connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the intake manifold pressure sensor from the engine harness.

Action	Specification/Repair	Next Step
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Inspect the engine harness and intake manifold pressure sensor connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	2B

STEP 2B. Check the sensor supply voltage and return circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the intake manifold pressure sensor from the engine harness.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Check the supply voltage and return circuit. Measure the voltage from the intake manifold pressure +5 VOLT SUPPLY pin to the intake manifold pressure RETURN pin at the sensor connector of the engine harness. Use a wiring diagram for pin identification and the following procedure for general multimeter usage. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	Between 4.75-VDC and 5.25-VDC? YES	2C
	Between 4.75-VDC and 5.25-VDC? NO	3A

STEP 2C. Check the circuit response.

Conditions : - Turn keyswitch OFF. - Disconnect the intake manifold pressure sensor from the engine harness. - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check the circuit response. - Place a jumper wire between the intake manifold pressure +5 VOLT SUPPLY pin and the intake manifold pressure SIGNAL pin at the intake manifold pressure sensor connector of the engine harness. Refer to the circuit diagram or the wiring diagram for connector pin identification. Check for the appropriate circuit response after 30 seconds. - Use INSITE™ electronic service tool to read the fault codes.	Fault Code 122 active and Fault Code 123 inactive? YES	2D
	Fault Code 122 active and Fault Code 123 inactive? NO	3B

STEP 2D. Check the fault codes and verify sensor condition.

Conditions :

- Turn keyswitch OFF.
- Connect the intake manifold pressure sensor to the engine harness.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 123 active? YES Repair : A damaged sensor has been detected. Replace the sensor. Refer to Procedure 019-159 in the Associated Procedures Table.	4A
	Fault Code 123 active? NO Repair : None. The removal and installation of the connector corrected the fault.	4A

STEP 3. Check the ECM and engine harness.

STEP 3A. Inspect the ECM and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM connector.

Action	Specification/Repair	Next Step
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Inspect the engine harness and ECM engine connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM engine connector or engine harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	3A-1

STEP 3A-1. Check the sensor supply voltage and return circuit.

Conditions : • Turn keyswitch OFF. • Disconnect the engine harness from the ECM connector. • Turn keyswitch ON.		
Action	Specification/Repair	Next Step

Check the supply voltage and return circuit. Measure the voltage from the intake manifold pressure +5 VOLT SUPPLY pin to the intake manifold pressure RETURN pin at the ECM engine connector. Use a wiring diagram for pin identification and the following procedure for general multimeter usage. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	Between 4.75-VDC and 5.25-VDC? YES	3A-2
	Between 4.75-VDC and 5.25-VDC? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A

STEP 3A-2. Check for an active fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 123 inactive? YES Repair : None. The removal and installation of the connector corrected the fault.	4A
	Fault Code 123 inactive? NO Repair : An open or shorted supply circuit has been detected in the engine harness. Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A

STEP 3B. Inspect the ECM and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM connector.

Action	Specification/Repair	Next Step

Inspect the engine harness and ECM engine connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM engine connector or engine harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	3B-1

STEP 3B-1. Check the ECM response.

Conditions : • Turn keyswitch OFF. • Disconnect the engine harness from the ECM connector. • Turn keyswitch ON. • Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check the ECM response. - Place a jumper wire between the intake manifold pressure +5 VOLT SUPPLY pin and the intake manifold pressure SIGNAL pin at the engine control module engine connector. Refer to the circuit diagram or the wiring diagram for connector pin identification. Check the appropriate circuit response after 30 seconds. - Use INSITE™ electronic service tool to read the fault codes.	Fault Code 122 active and Fault Code 123 inactive? YES	3B-2
	Fault Code 122 active and Fault Code 123 inactive? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A

STEP 3B-2. Check for an active fault code.

Conditions : - Connect all components. - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 123 inactive? YES Repair : None. The removal and installation of the connector corrected the fault.	4A
	Fault Code 123 inactive? NO Repair : An open circuit or a pin-to-pin short circuit has been detected on the SIGNAL wire of the engine harness. Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A

STEP 4. Check ECM calibration and clear fault codes.

STEP 4A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	4B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	4B

STEP 4B. Disable the fault code.

Conditions : - Connect all components. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete
	Fault code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/164/164-019-031.html)	ISL9.5 CM2150 SN	4310608
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/209/209-019-031-tr.html)	ISB3.9 CM2220 B107	4310792
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/218/218-019-031.html)	QSB4.5 CM2150 B108	4326163
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/216/216-019-031.html)	QSB6.7 CM2150 B109	4326168
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/331/331-019-031.html)	QSB7 CM2880 B117	4358390
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG11 CM2880 G106	4332695
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG12 CM2880 G107	4332690

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG11 CM2880 G108	4332901
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG12 CM2880 G109	4332906
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/206/206-019-031-tr.html)	QSNT14 CM876 N102	4325993
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F104	4332741
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F108	4332746
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/336/336-019-031.html)	ISL8.9 CM2880 L112	4358493
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/389/389-019-031-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/379/379-019-031.html)	ISB/ISD6.7 CM2880 B126	4383693

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/383/383-019-031-tr.html)	ISB5.9 CM2880 B127	4383645
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/387/387-019-031.html)	ISF3.8 CM2220 F116	4383664
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/403/403-019-031.html)	QSL9.3 CM2880 L113	4383811
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/436/436-019-031.html)	QSC8.3 CM2880 C102	4388785
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/451/451-019-031.html)	QSB5.9 CM2880 B139	4388870
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/449/449-019-031.html)	QSB3.9 CM2880 B138	5411050
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/506/506-019-031.html)	ISD6.7 CM2880 D101	5411372
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/497/497-019-031.html)	ISF2.8 CM2220 F129	5411325

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/501/501-019-031.html)	ISF4.5 CM2220 F123	5411320
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/417/417-019-031.html)	QSG12 CM2880 G112	4388731
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/556/556-019-031.html)	ISF3.8 CM2220 F134B	5504165
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/577/577-019-031.html)	Z14 CM2670 Z103B	5504577
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/546/546-019-031.html)	D6.7 CM2670 D102B	5504515
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/629/629-019-031.html)	B6.2 CM2670 B156B	5579510
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/624/624-019-031.html)	X12 CM2670 X121B	5504455
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/622/622-019-031.html)	L9 CM2670 L128B	5504589

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/164/164-019-043.html)	ISL9.5 CM2150 SN	4310608
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/209/209-019-043-tr.html)	ISB3.9 CM2220 B107	4310792
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/218/218-019-043.html)	QSB4.5 CM2150 B108	4326163
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/216/216-019-043.html)	QSB7 CM2150 B109	4326168
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/331/331-019-043.html)	QSB6.7 CM2880 B117	4358390
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/177/177-019-043-tr.html)	ISG11 CM2880 G106	4332695
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/177/177-019-043-tr.html)	ISG12 CM2880 G107	4332690
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/325/325-019-043-tr.html)	ISG11 CM2880 G108	4332901

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/325/325-019-043-tr.html)	ISG12 CM2880 G109	4332906
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/206/206-019-043-tr.html)	QSNT14 CM876 N102	4325993
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/284/284-019-043.html)	QSF2.8 CM2880 F104	4332741
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/284/284-019-043.html)	QSF2.8 CM2880 F108	4332746
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/336/336-019-043.html)	ISL8.9 CM2880 L112	4358493
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/389/389-019-043-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/379/379-019-043.html)	ISB/ISD6.7 CM2880 B126	4383693
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/383/383-019-043-tr.html)	ISB5.9 CM2880 B127	4383645

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/387/387-019-043.html)	ISF3.8 CM2220 F116	4383664
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/403/403-019-043.html)	QSL9.3 CM2880 L113	4383811
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/436/436-019-043.html)	QSC8.3 CM2880 C102	4388785
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/451/451-019-043.html)	QSB5.9 CM2880 B139	4388870
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/449/449-019-043.html)	QSB3.9 CM2880 B138	5411050
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/506/506-019-043.html)	ISD6.7 CM2880 D101	5411372
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/497/497-019-043.html)	ISF2.8 CM2220 F129	5411325
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/501/501-019-043.html)	ISF4.5 CM2220 F123	5411320

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/417/417-019-043.html)	QSG12 CM2880 G112	4388731
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/556/556-019-043.html)	ISF3.8 CM2220 F134B	5504165
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/577/577-019-043.html)	Z14 CM2670 Z103B	5504577
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/546/546-019-043.html)	D6.7 CM2670 D102B	5504515
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/629/629-019-043.html)	B6.2 CM2670 B156B	5579510
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/624/624-019-043.html)	X12 CM2670 X121B	5504455
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/622/622-019-043.html)	L9 CM2670 L128B	5504589
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/164/164-019-159.html)	ISL9.5 CM2150 SN	4310608

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/209/209-019-159-tr.html)	ISB3.9 CM2220 B107	4310792
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/218/218-019-159.html)	QSB4.5 CM2150 B108	4326163
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/216/216-019-159.html)	QSB6.7 CM2150 B109	4326168
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/331/331-019-159.html)	QSB7 CM2880 B117	4358390
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/177/177-019-159-tr.html)	ISG11 CM2880 G106	4332695
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/177/177-019-159-tr.html)	ISG12 CM2880 G107	4332690
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/325/325-019-159-tr.html)	ISG11 CM2880 G108	4332901
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/325/325-019-159-tr.html)	ISG12 CM2880 G109	4332906

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/206/206-019-159.html)	QSNT14 CM876 N102	4325993
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/284/284-019-159.html)	QSF2.8 CM2880 F104	4332741
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/284/284-019-159.html)	QSF2.8 CM2880 F108	4332746
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/336/336-019-159.html)	ISL8.9 CM2880 L112	4358493
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/389/389-019-159.html)	QSF3.8 CM2880 F112	4383825
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/379/379-019-159.html)	ISB/ISD6.7 CM2880 B126	4383693
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/383/383-019-159-tr.html)	ISB5.9 CM2880 B127	4383645
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/387/387-019-159.html)	ISF3.8 CM2220 F116	4383664

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/403/403-019-159.html)	QSL9.3 CM2880 L113	4383811
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/436/436-019-159.html)	QSC8.3 CM2880 C102	4388785
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/451/451-019-159.html)	QSB5.9 CM2880 B139	4388870
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/449/449-019-159.html)	QSB3.9 CM2880 B138	5411050
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/506/506-019-159.html)	ISD6.7 CM2880 D101	5411372
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/497/497-019-159.html)	ISF2.8 CM2220 F129	5411325
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/501/501-019-159.html)	ISF4.5 CM2220 F123	5411320
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/417/417-019-159.html)	QSG12 CM2880 G112	4388731

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/556/556-019-159.html)	ISF3.8 CM2220 F134B	5504165
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/577/577-019-159.html)	Z14 CM2670 Z103B	5504577
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/546/546-019-159.html)	D6.7 CM2670 D102B	5504515
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/629/629-019-159.html)	B6.2 CM2670 B156B	5579510
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/624/624-019-159.html)	X12 CM2670 X121B	5504455
Intake Manifold Pressure/Temperature Sensor	Refer to Procedure 019-159 (/qs3/pubsys2/xml/en/procedures/622/622-019-159.html)	L9 CM2670 L128B	5504589

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